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ABSTRACT

Boost converters are widely used in industry for many applications, such as electrical vehicles, wind energy systems, and photovoltaic energy systems, to step up the low voltages. Using the topology structure of the DC–DC boost circuit, this paper studied and designed a dual loop control method based on proportional integral controllers for improving the converter efficiency. The inner loop and outer loop controls of the traditional boost circuit are adopted in MATLAB/Simulink software to make the output of the system more stable. The input voltage is set to 24 V DC, and the desired output voltage varies from 36 to 48 V. Through simulation verification, the influence of a 1 kW sudden load connection by using a switch at a nominal output voltage of 48 V DC is studied, and the results show that it reduces the transient output voltage dips during the sudden load connection. Simulation analysis verifies the design scheme of the system, reduces the fluctuation in output voltage and power, reduces the output current ripple, minimizes the dip in voltage to a minimum possible value, and improves the dynamic characteristics and overall efficiency of the converter.

I. INTRODUCTION

With the continuous development of switching power supply technology nowadays, DC converters have a wide range of application in today's industry. They are commonly used in automotive industries, aircraft systems, hybrid electric cars, renewable energy power systems, and smart grids.[1] DC converters are of isolated and non-isolated types, and both have merit and demerits in terms of application. However, boost converters are an important part of the power supply technology due to their simplicity, and they have always been the focus of research by experts and scholars.[2,3] To improve the performance of these converters, it is essential to develop control strategies for them. The traditional controller inevitably increases the number of circuit magnetic components, resulting in an increase in the volume of the system and reducing the power density of the system.[4,5] The authors of Ref. 6 used discrete inductors to design interleaved parallel DC/DC converters, which made the system too large and the voltage and current ripples too high. The authors of Ref. 7 combined multiple converters to analyze the influence of different integration methods of coupled inductors on current ripples. The output current ripple and output voltage ripple are adjusted by changing the winding method of the coupled inductor. References 8 and 9 gave the equivalent circuit of coupled inductance characterized by mutual inductance.

The mathematical formula showed that the steady-state equivalent inductance increases during reverse coupling while the dynamic state decreases on the contrary, which conforms to the low steady-state ripple of the converter. In Refs. 10–12, the authors illustrated a closed loop controller for boost converters, in which the ripple in the inductor current decreases somehow but there is presence of dips in the output voltage.[13] Therefore, combining discrete inductors into coupled inductors in the interleaved parallel boost circuit can not only reduce the volume of the converter but also increase the power density and reduce the output current ripple with a suitable coupling coefficient.[14] In Refs. 15 and 16, a voltage and current double closed loop control method for the boost circuit is introduced, which determined the coupling inductance of the interleaved parallel boost circuit under the condition of a large duty cycle. However, it improved the circuit output, the voltage ripple became smaller, and it has better system robustness and faster dynamic response.[17,18] In order to overcome the shortcomings in the traditional controller of a boost converter, such as to reduce the current ripple of the converter, increase the power density of the system, reduce dips in the output voltage, and increase the overall efficiency, as well as reduce the transient output voltage dips during a sudden connection of the load, this paper proposed a dual loop controller design based on PI control topology to meet the requirements.

The main contribution of this paper is the introduction of a novel dual loop control method, utilizing a proportional integral (PI) controller, to improve the efficiency of boost converters. These converters are widely used in industrial applications such as electric cars, wind energy systems, and photovoltaic energy systems. This research undertakes a comprehensive examination and design investigation centered on the topological configuration of the DC–DC boost circuit. This demonstrates a realistic comprehension of the boost converter and its operational concepts. This work presents an in-depth and practical method to improve the efficiency and stability of boost converters. It offers significant insights into the dynamic behavior of the system when subjected to unexpected load fluctuations. The utilization of the dual loop control approach and its effective implementation significantly enhance the progress of control strategies for DC–DC boost circuits.

In this article, Sec. II describes the boost converter analysis and designs the mathematical equations of the boost converter. Based on the mathematical equations, the parameters of the boost converter are calculated. The mathematical equations for calculating the gain signal are also described. Section III is divided into circuit discussion and simulation results. All the three cases are elaborated in detail. In case 1, the boost converter uses a dual loop controller; in case 2, a step is implemented to get the desired variable output voltage; in case 3, a sudden load is connected to the system. In all three cases, the designed dual loop controller performs efficiently.

II. BOOST CONVERTER ANALYSIS

A DC–DC converter is also known as a DC chopper. By regulating the switch, it converts the uninterrupted DC voltage into an adjustable or fixed DC voltage. DC–DC converters have many categories, generally based on the voltage and current waveforms output by power switches.[19,20]

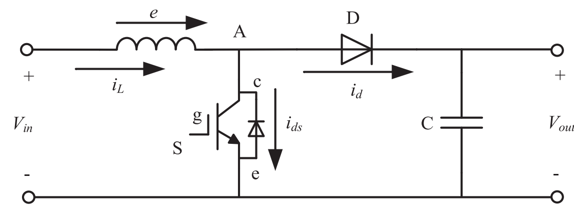

FIG. 1. Boost converter circuit.

The work process of boost converter transformation is as shown in Fig. 1, which shows the equivalent circuit diagram of the boost converter.[21] Under the T_{on} condition of MOSFET transistor switch S, at which point the power supply will only provide electrical energy to inductor L, and before the inductor reaches saturation, the electrical energy will be converted into electromagnetic energy, which is stored in inductor L. At this time, inductor L has positive voltage at both ends, inductive current i will increase, and the load of the energy supply will be provided by capacitor C. Flowing through the load, the voltage at both ends of the load is positive and negative, in T_{on} ~ 0. At this high-level trigger, the switch S is in the on state, and the diode D is in the cut-off state under the action of the antibias voltage because capacitor C is not discharged by the switch.[22,31] When the switch S is in the cut-off state, inductor L is under the action of magnetic field, the voltage polarity of both ends of the inductor will change, the inductive voltage and supply voltage stored by magnetic energy conversion will be grouped into a series voltage, and the formation of the series voltage will be greater than the input supply voltage. This series voltage provides electrical energy for capacitor C and load R, which flowing into the charging current. When there is a decreasing tendency, it provides power to load R so that the voltage remains the same. Hence, the output voltage V_o will always be high at the supply voltage, so the converter is called a boost converter.[23,24]

The circuit energy conservation relationship is obtained as

$$V_i * T_{on} = (V_o - V_i) * T_{off}, \tag{1}$$

where V_i = input voltage and V_o = output voltage across the load. The circuit duty-free ratio D is

$$D = \frac{T_{on}}{T_{on} + T_{off}}. \tag{2}$$

When the boost transformation circuit is stable, the output voltage mean is

$$V_o = \frac{T_{on}}{T_{on} + T_{off}} V_i = \frac{T}{T_{off}} = \frac{1}{1 - D} V_i. \tag{3}$$

Thus, the output of the boost transform circuit is derived from the input voltage relationship,

$$\frac{V_o}{V_i} = \frac{1}{1 - D}. \tag{4}$$

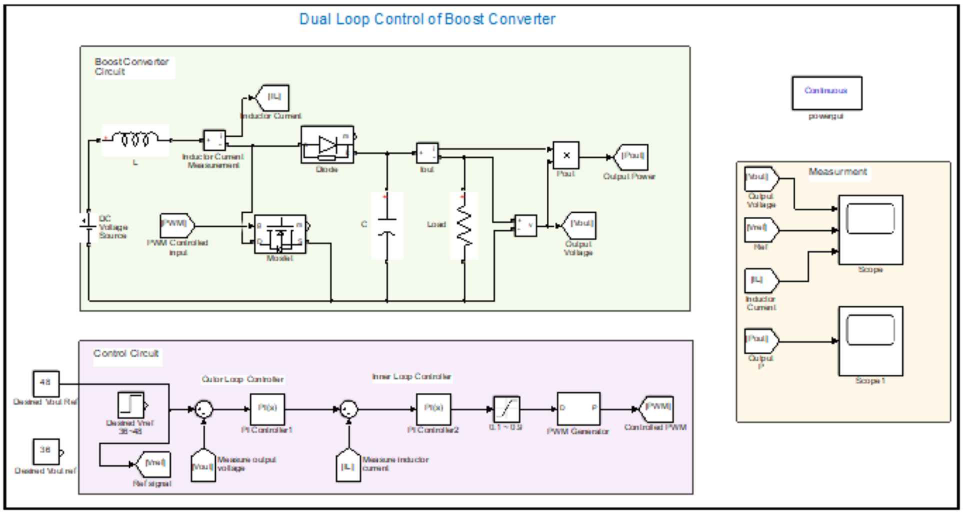

FIG. 2. Dual loop control circuit of the boost converter.

The inductor current in the boost converter consists of two parts, direct flow and ripple components, where the ripple current components can be expressed as

$$\Delta I = \left(\frac{V_i}{L}\right) T_{on} = \left(\frac{V_i}{L}\right) DT. \tag{5}$$

Ignoring the internal losses of the circuit yields

$$V_i I_i = V_o I_o \tag{6}$$

$$L = \frac{V_i T_{on}}{1.4 I_i} = \frac{Vo^2 (V_o - V_i)}{1.4 f\, Vo^2 I_o}, \tag{7}$$

$$C = \frac{I_o T_{on}}{\Delta V_o} = \frac{V_o DT}{\Delta V_o R} = \frac{V_o D}{\Delta V_o R f}. \tag{8}$$

A. Regulatory technology of the DC/DC transform circuit

The regulation technology of a DC/DC transform circuit contains two aspects: one is the switch control strategy, which is commonly referred to as the tuning technology; the other is the closed-loop control strategy and dual loop control strategy of the circuit, which is commonly referred to as the control technology. Boost circuit modulation technology includes pulse width modulation (PWM),[11] Pulse Frequency Modulation (PFM),[25] and Pulse Density Modulation (PDM).[26] In new energy generation systems, PWM technology is generally chosen, and the main principle is to maintain the switching cycle unchanged. In PWM technology, the front, back, or front and rear edges of a pulse change with the modulation signal. The carrier frequency is fixed in PWM technology. It is easy to select devices and filter the parameter design, its input and output linearity is good, transmission bandwidth is high, and it is widely used in all kinds of electrical and electronic transformation circuits.[27]

In this article, the control technology of a DC–DC boost circuit mainly refers to its dual-loop control strategy. From the point of view of control requirements, it can be divided into voltage type control, current control, and dual loop control. From the composition of a closed-loop regulator, it can be divided into PI control, hysteresis control, and single-cycle control. In the new energy generation system, because of its hardware structure, the dual loop PI controller-based control strategy is mainly used.[28–31]

The main principle of dual loop control strategy of a boost converter is to distinguish between a given voltage V_{ref} and the output voltage V_{out}. Through the PI regulator, it is shaped into a voltage modulation signal and then the voltage modulation signal and fixed frequency wave signal's intersection is used to form a pulse signal. When a given voltage, load, or power supply changes, the intersection of the modulated signal and the jagged wave changes, changing the pulse width to achieve the role of voltage dual-loop control,[24]

$$\mathrm{u\_v(t) = Kpv \times e\_v(t) + kiv \times \int e\_v(t)\, dt.} \tag{9}$$

TABLE I. Case one and two parameter values.

Parameters	Values
Input voltage	24 V DC
Desired output voltage	36, 48, and 36–48 V
Inductor	1100 μH
Capacitor	500 μF
Switching frequency	125 kHz
Delay angle	0.1–0.9
Kp	0.4 for both inner and outer loops
Ki	80 for both inner and outer loops

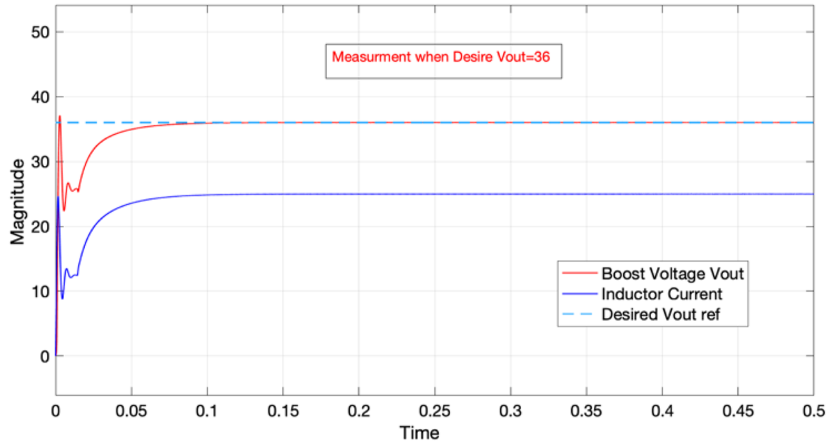

FIG. 3. Waveform of boost voltage and inductor current (Vref = 36 V).

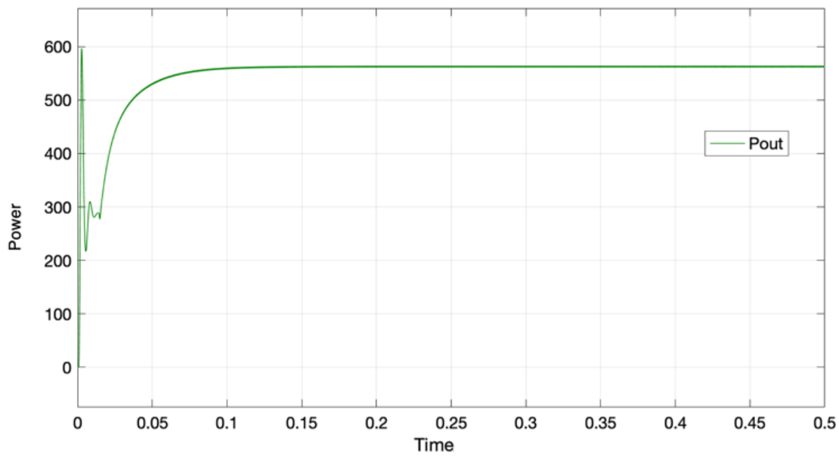

FIG. 4. Waveform of output power (Vref = 36 V).

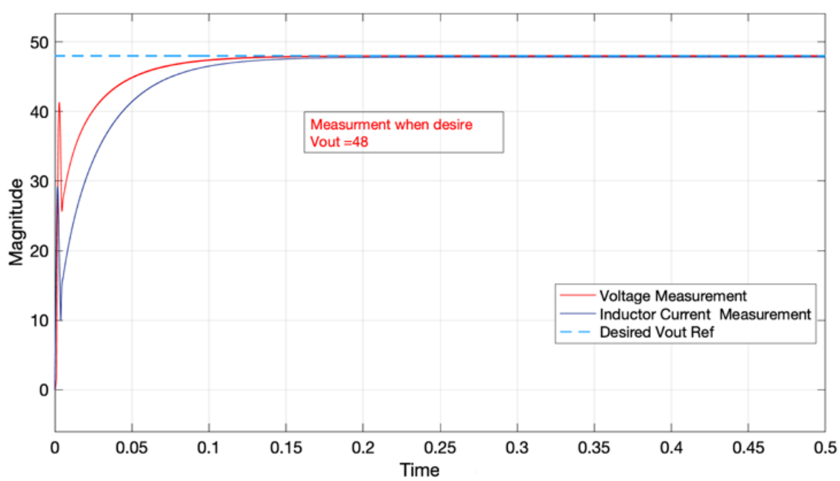

FIG. 5. Waveform of boost voltage and inductor current (Vref = 48 V).

The equation given above illustrates the outer voltage loop controller, where u_v(t) is the voltage loop control signal, Kpv is the proportional gain, and the integral gain is Kiv. The outer loop controller is used to regulate the output voltage,

$$u\_i(t) = Kpi \times e\_i(t) + kii \times \int e\_i(t)\,dt, \tag{10}$$

where u_i(t) is the current loop control signal, Kpi is the proportional gain, and the kii is the integral gain for current control.

The control signal for the boost converter is calculated by the following equation:

$$u(t) = u\_v(t) + u\_i(t), \tag{11}$$

$$D(t) = u(t)/Vdc, \tag{12}$$

where D is the duty cycle and Vdc is the source input voltage. The duty cycle of the boost converter is determined from this controller by using the PWM generator that is shown in Fig. 2. Both gain

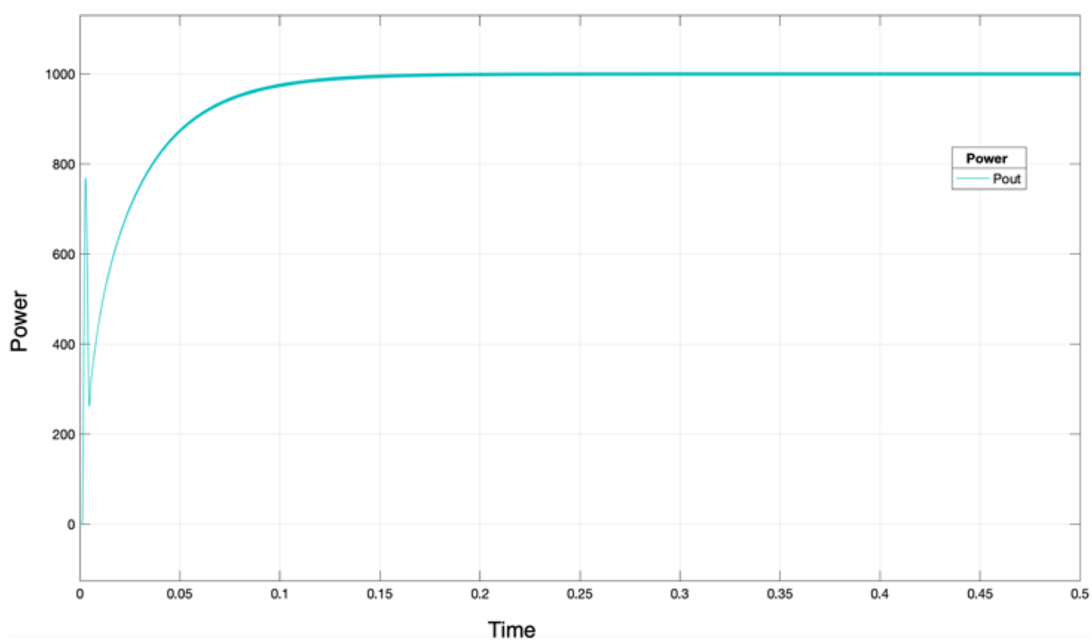

FIG. 6. Waveform of output power (Vref = 48 V).

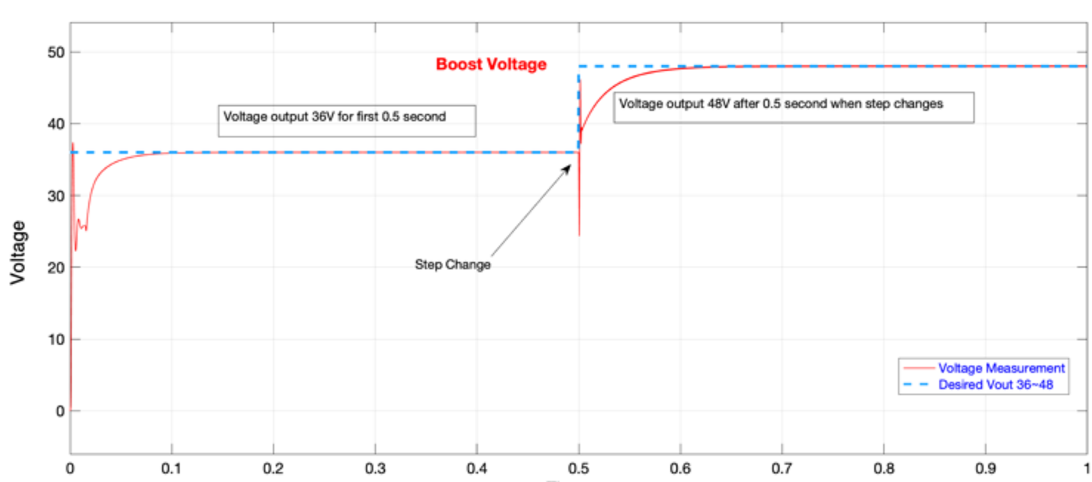

FIG. 7. Output boost voltage during step change.

signals Kp and Ki are properly tuned for the requirement of the system, and the values for both gains are listed in the tables given in the Simulation and Results section.

III. SIMULATION AND RESULTS

A. Circuit analysis

To further verify the above-mentioned theoretical analysis, simulation research is carried out in a MATLAB/Simulink platform. Mainly the circuit consists of a controller portion (inner controller and outer controller) and a DC–DC boost converter circuit. The dual loop controller is a closed loop, and this system uses two feedback loops to regulate a process, as well as to enhance the control accuracy and performance by controlling both primary and secondary variables. In addition, two PI controllers are used; one of these is for the inner loop controller, and the second is for the outer loop controller. The controlled signal of the PI controller is provided to the boost converter through the DC–DC PWM generator. A variable delay is provided from 0.1 (lower value) to 0.9 (higher value). However, the measurement section consists of a scope and digital measurement display to monitor the input and output parameters of the circuit. The circuit index is input side voltage Vin = 24 V DC, output side variable voltage = 36–48 V DC,

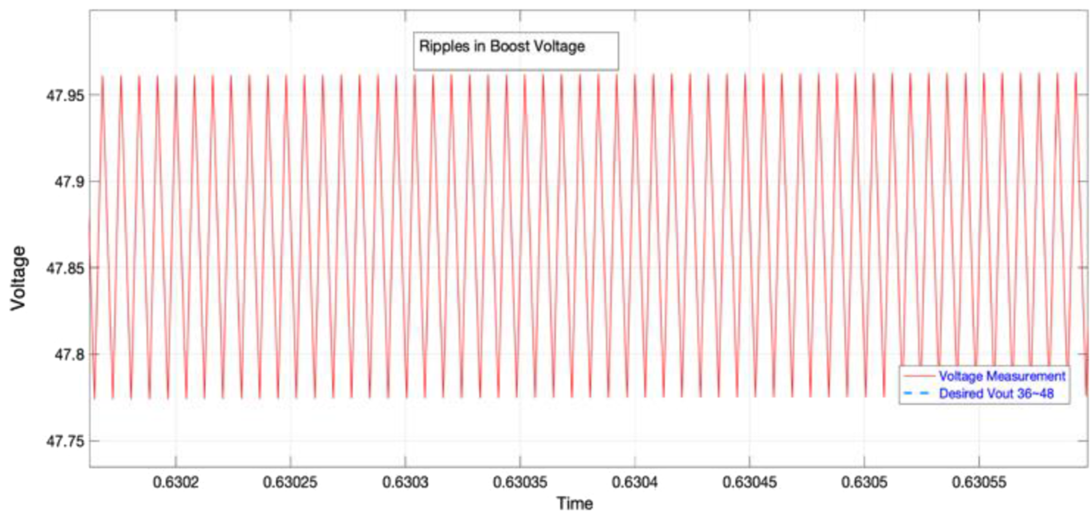

FIG. 8. Ripples in boost voltage.

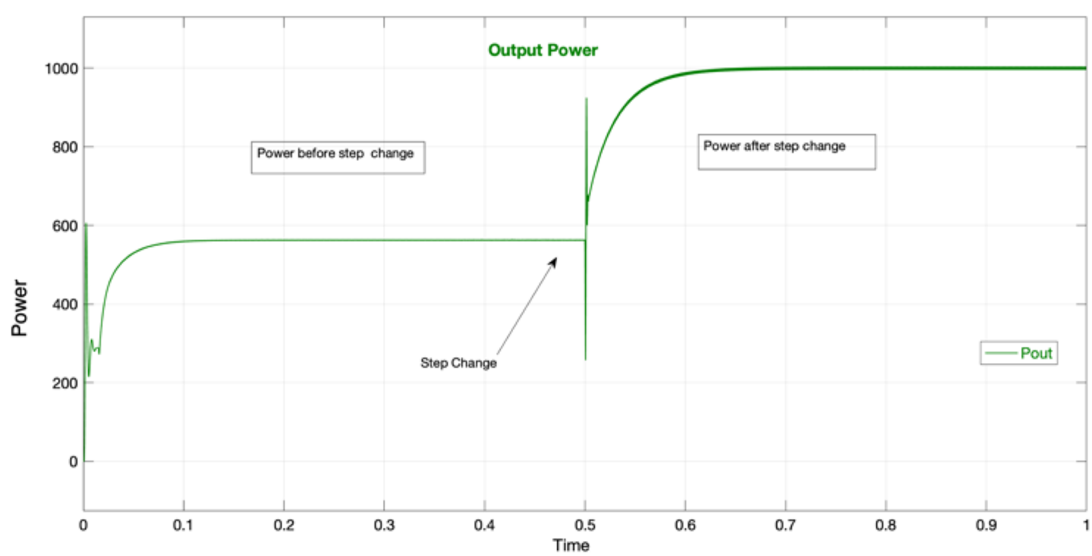

FIG. 9. Output power during step change.

and operating frequency f = 125 kHz. Based on the mathematical equations discussed in Sec. II, the circuit parameters are given as follows: inductance L = 1100 μH, filter capacitor C = 500 μF, the power switch chooses a power MOSFET, and output load = 1 kW. The inner and outer dual loop controllers are designed according to the transfer function of the system, and the parameters are adjusted in the actual adjustment process. The inner loop and outer loop consist of a PI controller. By using a step function, we can get two different desired output voltages; the step time is set at 0.5 s, which indicates that after 0.5 s, the desired output voltage increases to 48 V. The circuit diagram of the dual loop control of the boost converter based on the proportional integral PI controller is shown in Fig. 2.

B. Simulation results

Three different scenarios of a dual loop control system are described. The first one takes into account a fixed desired output voltage, the second introduces step changes in the reference voltage from 36–48, and the third scenario is the sudden load connection by using an ideal switch.

Case One: The nominal values of the boost converter parameters are given in Table I. In this case, the desired output voltage is 36 V DC, and the simulation time is set at 0.5 s. Figure 3 illustrates the refence voltage signal, inductor current, and output boost voltage. The value of the inductor current is recorded at about 24 A, and the output boost voltage is recorded as 36 V, which can be

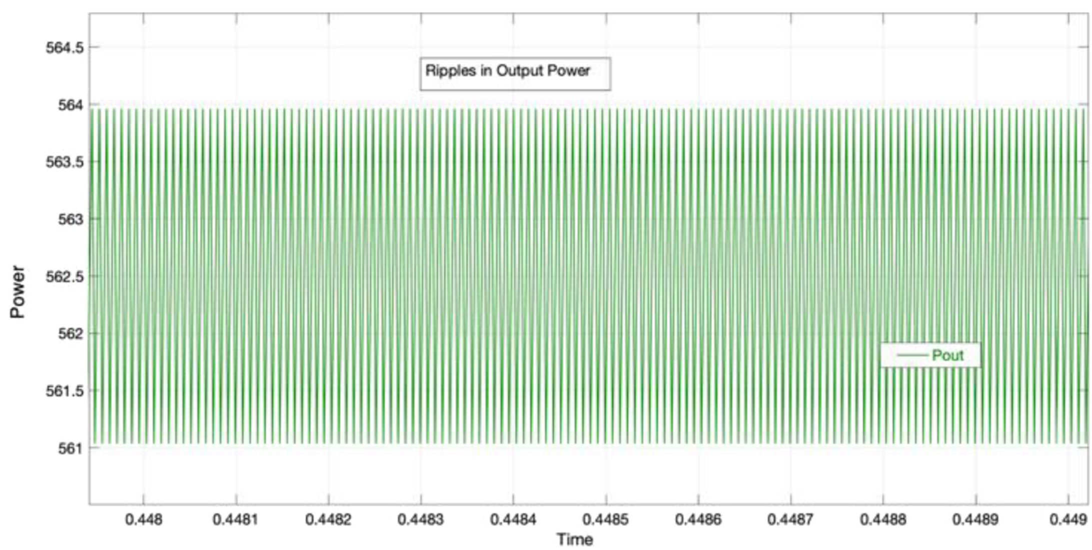

FIG. 10. Ripples in output power.

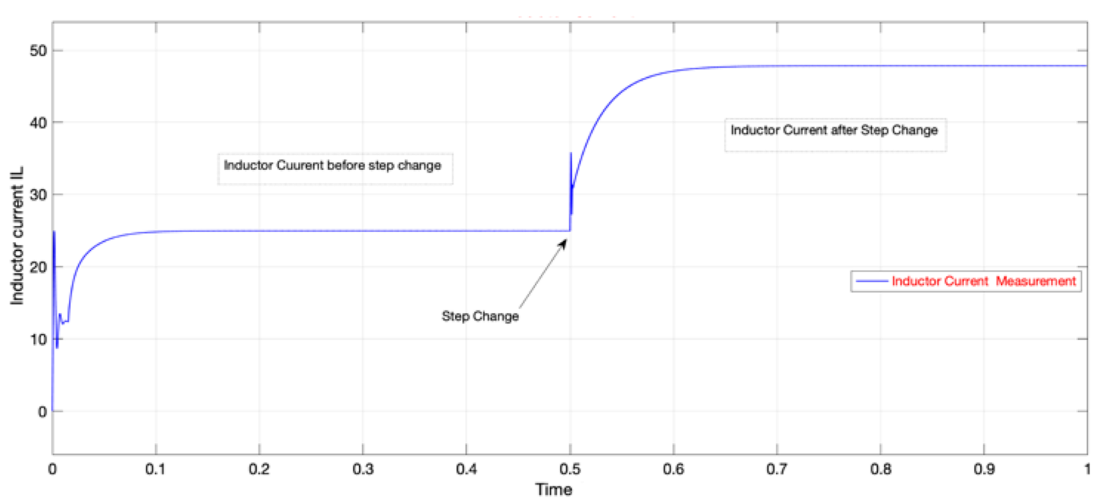

FIG. 11. Inductor current during step change.

seen in the waveform shown in Fig. 3. The dip in the voltage reduces to a minimum possible value. It can be seen from the figure that after a slight dip at 0.05 s, the output value becomes stable. However, in this case, the output power is recorded at 570 W, which is shown in Fig. 4.

The efficiency of the converter in this case is about 95%. However, when the desired output voltage value is changed to 48 V DC, the output waveform of the inductor current and boost voltage is as shown in Fig. 5, while the output power is as shown in Fig. 6. It can be analyzed from the results that the output boost voltage is 48 V. The inductor current increases to 48 A, and the output power is recorded as 1000 W.

Case two: In this scenario, the nominal values of the boost converter parameters are the same, but we get variable output

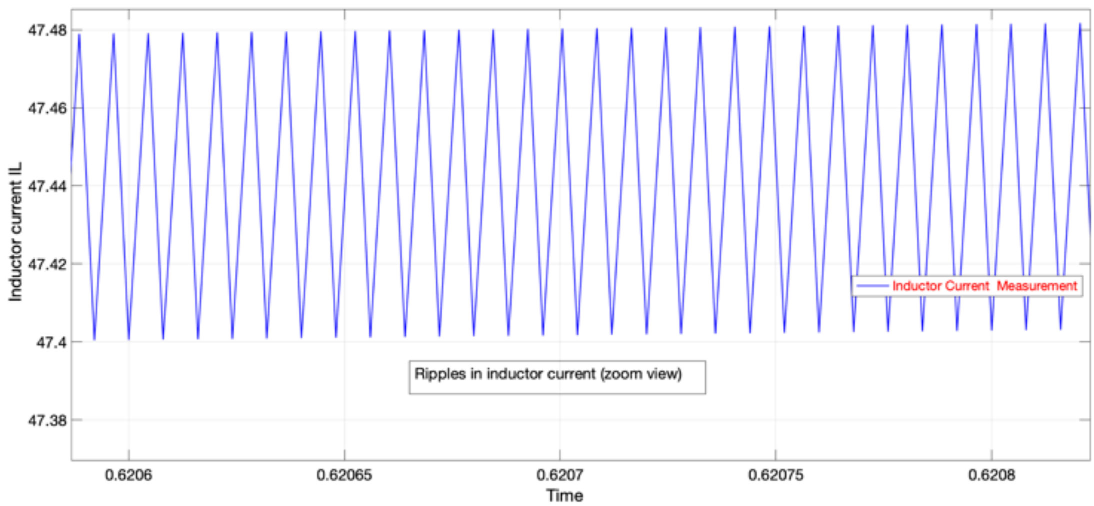

FIG. 12. Ripples in inductor current during step change.

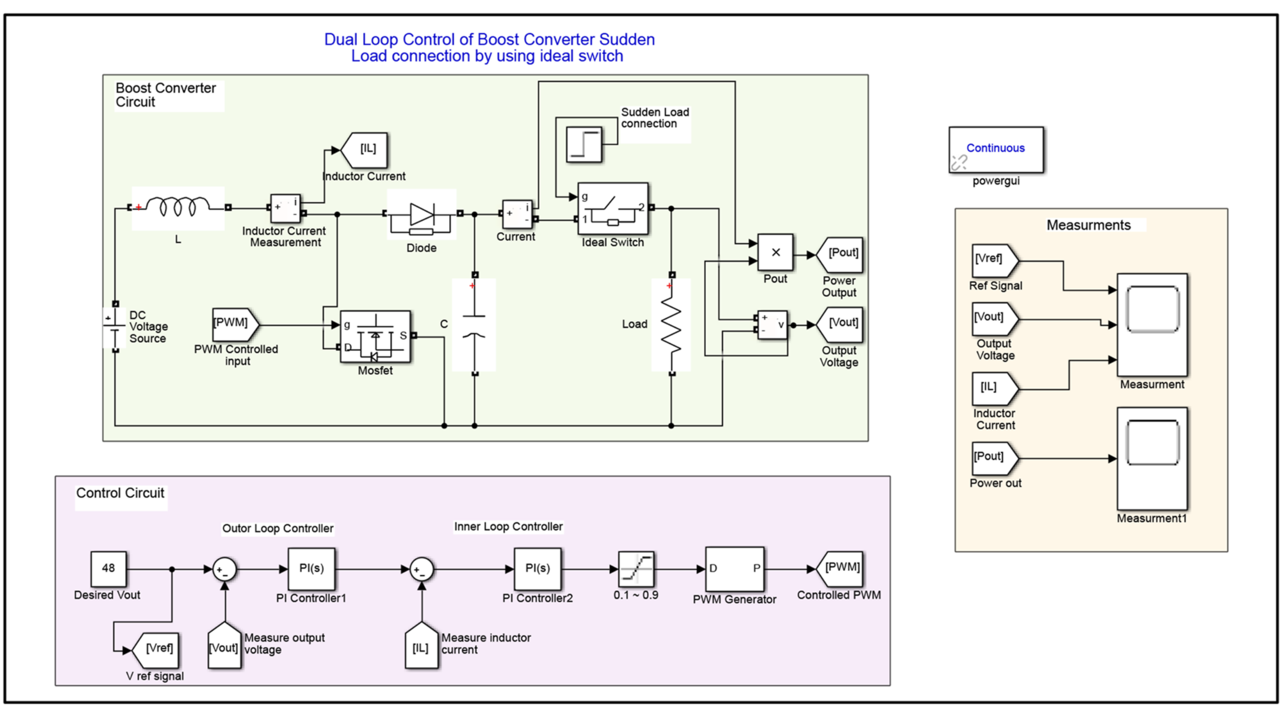

FIG. 13. Dual loop control of the boost converter by adding 1 kW sudden load at the output.

TABLE II. Case three parameter values.

Parameters	Values	
Input voltage	24 V DC	
Desired output voltage	48 V DC at 1 kW	
Inductor	1000 μH	
Capacitor	1500 μF	
Switching frequency	100 kHz	
Delay angle	0.1–0.9	
Kp	Inner loop 0.758 29	Outer loop −0.8
Ki	Inner loop 2	Outer loop 8

voltage in the range 36–48 V DC by using step reference. The parameters of step reference are set to step time = 0.5, initial value = 36, and final value = 48. In this case, the simulation time is set at 1 s. Figure 7 shows the behavior of the boost output voltage during the change in output voltage regulation. It can be observed from the waveform that during the initial time (first period for 0.5 s), the desired output voltage of 36 V is achieved successfully, while after step-changing, the system works perfectly and 48 V is achieved; in addition, the dip is reduced to a possible value that can be ignored. However, there is slight fluctuation recorded in the output voltage (0.05–0.2 V), which can be seen in Fig. 8.

Figure 9 shows the power output waveform during step changes in the desired output voltage. After 0.05 sec, the output power become stable and it reaches a maximum value of 580 W for the first half cycle. As the desired output voltage value increases to 48 V, the output power in the second half cycle increases to 1000 W successfully. Under this condition, there is slight fluctuation recorded in the output power (1–5 W), as shown in Fig. 10.

Figure 11 illustrates the inductor current during step change to regulate the output voltage of the boost converter. It can be seen from the waveform that during the first cycle (0.5 s), the inductor current is recorded at 24 A, and after step change, the inductor current increased to 48 A. However, negligible ripples of less than 0.3 A are recorded in the inductor current, as shown in Fig. 12.

Case Three: In this scenario, the nominal values of the boost converter parameters are slightly changed, and an extra ideal switch is employed to the circuit for sudden load connection, which is shown in Fig. 13. Parameters of the circuit can be seen in Table II. In this scenario, the desired voltage is set at 48 V, the sudden load connected is 1 kW, the time of sudden load connection by using an ideal switch is 0.1 s, and the simulation time is set at 1 s. From the period 0.1–0.9, it is seen from the waveform that after load connection, the desired output voltage is achieved successfully and the dip in voltage and inductor current reduces to a minimum value. Figure 14 illustrates that when the load is suddenly connected by using an ideal switch, a circuit step is employed to the gate of the switch and the parameters are set to step time = 0.1, initial value = 0, and final value = 1.

It can be observed from Fig. 14 that when 1 kW sudden load is connected to the boost converter after 0.1 s of operating time, the system works efficiently, and the desired output voltage of 48 V is obtained at 0.5 s. During sudden load connection, transient output voltage is negligible and the efficiency of the system is at about 98%. Figure 15 shows the slight ripples in the output voltage and inductor current, and the output power is illustrated in Fig. 16. In addition, it can be observed from the results that as the desired output voltage increases, the boost converter works efficiently under a load of 1 kW and the dynamic response of the converter is improved.

In addition, this dual loop control strategy for the boost converter provides variable output voltages based on the control of the delay angle of the switch. However, it efficiently works when a sudden load is connected to the system.

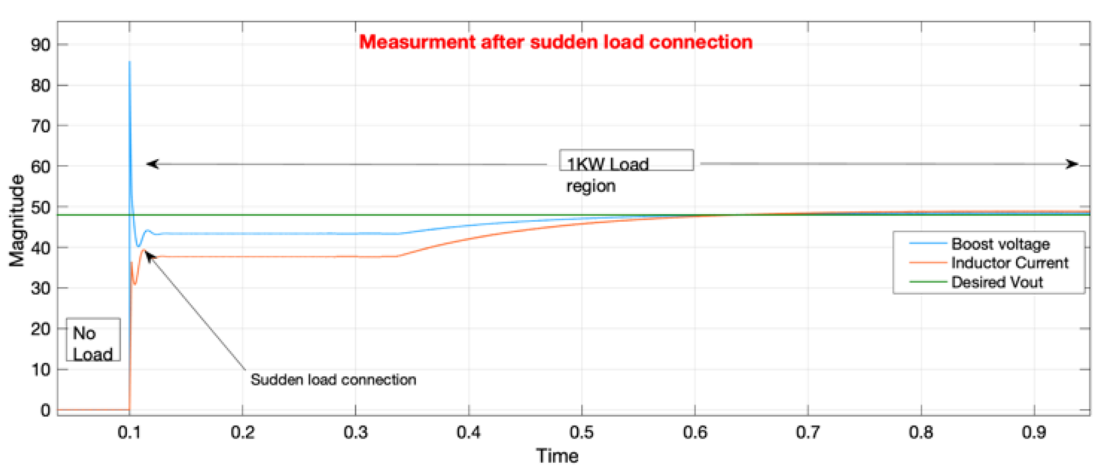

FIG. 14. System behavior after connecting a sudden load of 1 kW.

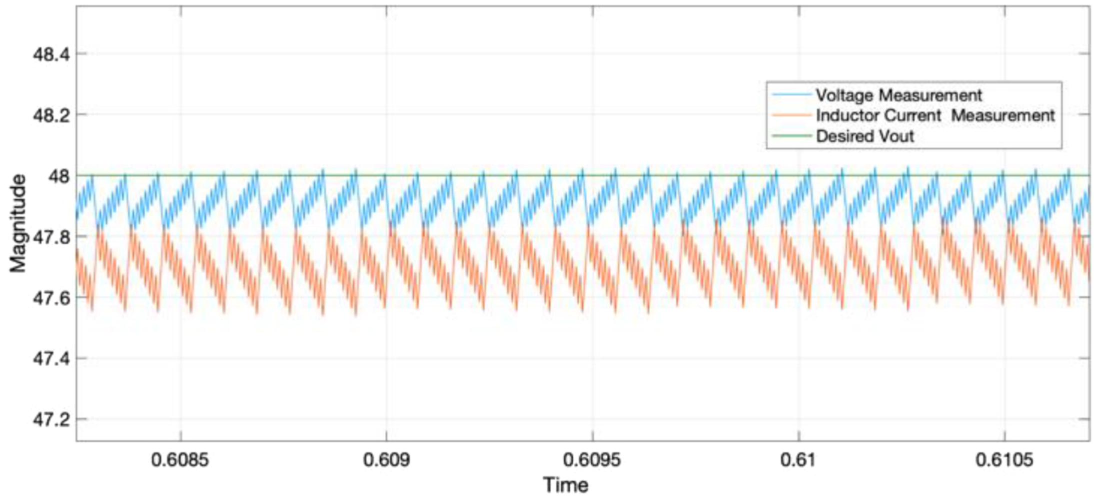

FIG. 15. Minimum transient waveform.

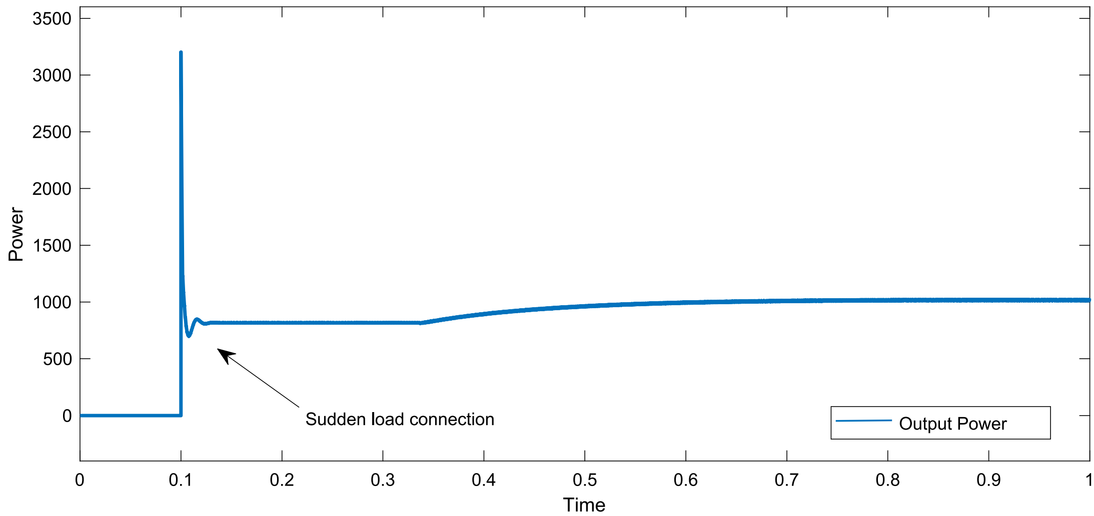

FIG. 16. Output power.

IV. CONCLUSION

The aim of this simulation design is to observe the regulated output voltage by using a dual loop control approach based on the PI controller. Two PI controllers are used: one is an inner loop controller, and the second is an outer loop controller. Detailed analysis of the boost circuit is described by mathematical equations, and analysis of the influence of switching frequency and load on the dynamic and steady-state characteristics of the circuit is carried out. Three scenarios are discussed in this article, such as constant desired output voltage, dynamic output voltage, which varies from 36 to 48 V, and the behavior of the converter when a sudden load is connected by using an ideal switch at nominal 48 V connection. The simulation results show that the inductor current ripples and transient reduce to a minimum possible value. In addition, the output voltage ripple becomes smaller, and the dynamic and steady state characteristics are improved. However, multiple output voltages can be achieved from the same converter by adjusting the delay angle of the converter switch. In addition, the overall efficiency of the converter is also improved.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Ethics Approval

This study does not involve any ethical issues, and data collection was completed in accordance with the ethical regulations.

Author Contributions

Kifayat Ullah: Conceptualization (equal); Methodology (equal); Software (equal); Writing – original draft (equal). **Muhammad Ishaq**: Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Validation (equal); Visualization (equal); Writing – review & editing (equal). **Muhammad Ajmal Naz**: Investigation (equal); Resources (equal); Validation (equal); Writing – review & editing (equal). **Mostafizur Rahaman**: Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Project administration (equal); Resources (equal); Writing – review & editing (equal). **Arsalan Muhammad Soomar**: Conceptualization (equal); Methodology (equal); Validation (equal); Writing – review & editing (equal). **Hijaz Ahmad**: Data curation (equal); Investigation (equal); Supervision (equal); Writing – review & editing (equal). **Md. Nur Alam**: Conceptualization (equal); Formal analysis (equal); Methodology (equal); Validation (equal); Visualization (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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